Descriptive Statistics

30/9/25

Mean, Variance, Moments, Centralized moments

(expectation) $E[X]$ $E[(X-E[X])^2]$ $E[X^n]$ $E[(X-E[X])^n]$

$n=1$: 0

$n=2$: variance

$X_1, X_2, \dots, X_n$: joint distribution

$$\mathrm{Var}(X_1+X_2+\cdots+X_n) = E[(X_1+X_2+\cdots+X_n)^2] - (E[X_1]+\cdots+E[X_n])^2$$

$$= E[X_1^2]+\cdots+E[X_n^2] + \sum_{\substack{i=1\\ i\neq j}}^{n}\sum_{j=1}^{n} E[X_i X_j] - E[X_1]^2 - \cdots - E[X_n]^2 - \sum_{\substack{i=1\\ i\neq j}}^{n}\sum_{i=1}^{n} E[X_i]E[X_j]$$

$$= \mathrm{Var}(X_1)+\cdots+\mathrm{Var}(X_n) + \sum_{i\neq j}\sum \left(E[X_iX_j]-E[X_i]E[X_j]\right)$$

$n=2$

$$\mathrm{Var}(X_1+X_2) = \mathrm{Var}(X_1)+\mathrm{Var}(X_2)+2\left(E[X_1X_2]-E[X_1]E[X_2]\right)$$

"correlation between X_1 & X_2"

Correlation between X & Y : $E[(X-E[X])(Y-E[Y])]$

"$\mathrm{Cov}(X,Y)$"

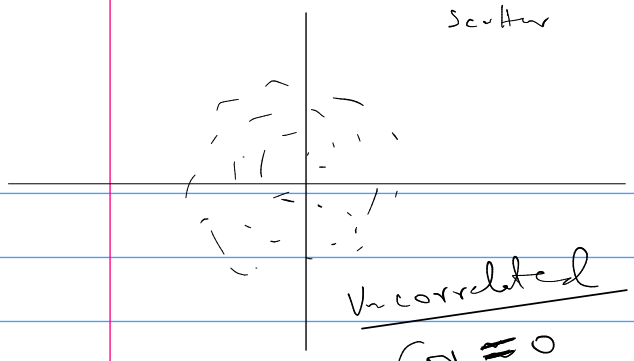
Covariance

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$ Exercise

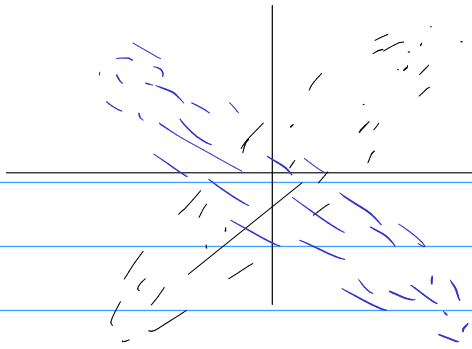
$E[XY]$ correlation

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y)$$

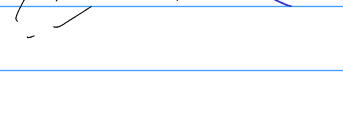
X, Y indep: $\mathrm{Cov}(X,Y)=0$



Uncorrelated
$Cov = 0$



high +ve correlation
ρ_{XY} : close to 1



high −ve Correlation
ρ_{XY} : close to −1

→ X, Y are "uncorrelated" if $Cov(X,Y) = 0$

Independence $\Rightarrow$ Uncorrelated (in general $\not\Leftarrow$)

Cauchy–Schwarz inequality

$x = (x_1, \dots, x_n) \in \mathbb{R}^n$

$y \in (y_1, \dots, y_n) \in \mathbb{R}^n$

$$|\langle x, y \rangle| \le \|x\| \cdot \|y\|$$

$\langle x, y \rangle$: $\sum x_i y_i$

$\|x\|$: $\sqrt{x_1^2 + \dots + x_n^2}$

$\|y\|$: $\sqrt{y_1^2 + \dots + y_n^2}$

equality iff $y = cx$ for some $c \in \mathbb{R}$

U, V: RVs

$$E[UV]^2 \le E[U^2] \cdot E[V^2]$$

equality iff $aU + bV = 0$ for some $a, b \in \mathbb{R}$.
$(P(aU + bV = 0) = 1)$

Pf:

$$0 \le E[(aU + bV)^2] = a^2 E[U^2] + b^2 E[V^2] + 2ab E[UV]$$

$a = \sqrt{E[V^2]}$

→ $b = \sqrt{E[U^2]}$ $\Rightarrow E[UV] \ge -\sqrt{E[U^2]E[V^2]}$

→ $b = -\sqrt{E[U^2]}$ $\Rightarrow E[UV] \le \sqrt{E[U^2]E[V^2]}$

$\blacksquare$

$$-\sqrt{E[U^2]E[V^2]} \le E[UV] \le +\sqrt{E[U^2]E[V^2]}$$

$$Cov(X,Y) = E\{\underbrace{(X-E[X])}_{U}\underbrace{(Y-E[Y])}_{V}\}$$

Correlation Coefficient

$$\rho_{XY} = \frac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

$Var(X) = \sigma_X^2$

$Var(Y) = \sigma_Y^2$

$$-1 \leq \rho_{XY} \leq +1$$

equality iff $aX+bY=0$ line

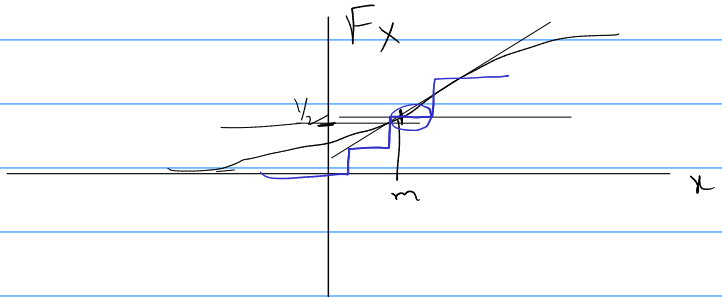
Median: $med(x_1, x_2, \ldots, x_n) = ?$

X: RV

$med(X)$:

m is a median for X if

$Pr(X \geq m) \geq 1/2$ and $Pr(X \leq m) \leq 1/2$



Prediction using mean & median

X: RV Find d that will predict 'X'.

d: constant

$E[|X-d|]$ — Mean of absolute deviation

$d = med(X)$ minimizes

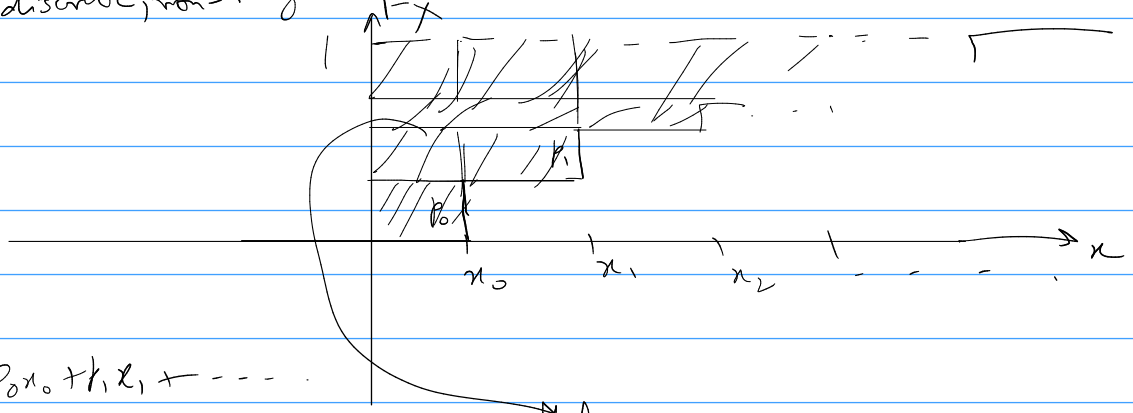
$E[(X-d)^2]$ — quadratic in d

mean square error (MSE)

$d = E[X]$ minimizes MSE

Expectation from CDF

→ discrete, non-negative



$E[X] = p_0 x_0 + p_1 x_1 + \cdots$

Area

$$E[X] = \int_0^\infty (1 - F_X(x))\,dx$$

→ holds for any non-negative RV.